

New Feature Overview Document:

Plant functional type definition and physiological changes for improved arctic vegetation representation

Feature description:

These code changes enable more flexible definitions of plant functional types (PFTs) in ELM and add vegetation physiological functions to better represent arctic vegetation: Prior to these changes, ELM assumes a specific set of PFTs, with hard-wired PFT name list, orders, and blocks. The code changes in this pull request allow the number of PFTs to be defined in the model configuration, and remove requirements that PFTs be defined in a particular order or grouped into specific blocks.

The physiological changes include multiple new functionalities relevant to defining arctic plants. First, changes are introduced to allow more flexible use of evergreen plants. In the current model, nonwoody plants are assumed to be deciduous. However, many arctic graminoids have fine roots and leaves that persist over winter. To enable evergreen graminoids, we introduce an option for reduced maintenance respiration for fine roots when soil temperatures are below a threshold defined in the plant physiology parameters. We also introduce code changes allowing nonwoody graminoid plants to allocate biomass to rhizomes, which are represented using the live coarse root pool. This allows better consistency with arctic graminoid growth forms, which maintain significant biomass in longer-living rhizome tissues that have distinct stoichiometry and residence times compared to fine roots. In addition, we introduce changes allowing evergreen plants to transfer C, N, and P from storage pools to growth. The current model assumes that the *f_{cur}* parameter (fraction of photosynthesis that is immediately allocated to growth) will be 1.0 for evergreen plants, assuming that evergreen plants do not use storage pools for C, N, or P, and the current model code does not transfer any C, N, or P from storage to growth for evergreen plants. We introduce new code that allows these transfers when evergreen plants do accumulate storage pools. This allows arctic plants to maintain storage pools over multiple years, which is important for defining PFTs that can survive in a climate with short growing seasons. We also introduce a change to phenology that allows deciduous plants to begin leaf-out after the summer solstice, to allow plant growth in cases when high accumulation of snow leads to late onset of phenological spring in high-latitude ecosystems. Finally, we introduce changes to the nitrogen fixation functionality to allow nitrogen fixation parameters to be defined with PFT-specific values. This allows the definition of nitrogen-fixing PFTs, which are important for representing the impact of nitrogen-fixing alder shrubs on nitrogen cycling in arctic ecosystems.

Detailed description of code changes:

Flexible PFT definitions:

The following new PFT properties or flags are introduced, to replace assumptions related to ordering or blocks of PFTs:

Lifeforms:

(1) nonvascular – 0: vascular/bare-ground, 1: moss, or 2: lichen.

(2) woody – 0: non-woody, 1: tree, or 2: shrub

(3) graminoid – 0: non-graminoid, or 1: graminoid (crops excluded)

(4) iscft – 0: non-crop-function-type (incl. non-crop, and generic crop), or 1: crop-functional-type (i.e. prognostic crop, including both non-generic crop and perennial crop)

Additional flags:

(1) climatezone – 0: non-specific, 1: tropical, 2: temperate, 3: boreal, or 4: arctic. (Needed to support fire calculations which previously depended on PFT ordering)

(5) needleleaf – 0: broadleaf, or 1: needleleaf

(6) nfixer – 0: non-nfixer, or 1: nfixer, i.e. able to fix N from atm. N₂

Compared to the original PFT physiology, the previously binary ‘woody’ (0 or 1) is changed to distinguish between nonwoody (0), tree (1) and shrub (2). Lifeform ‘nonvascular’ is a placeholder for moss or lichen, such as in boreal, arctic, etc. Flag ‘needleleaf’, together with existing flags of ‘evergreen’, ‘season_decid’, or ‘stress_decid’, are enough to distinguish among the existing tree and shrub PFTs for model functions that are specific to those existing PFTs. Flag ‘graminoid’ is for non-crop grass only.

The existing flag for crop PFTs, ‘crop’, applies to both generic and prognostic crops, and the existing code depends on identifying the two existing generic crop PFTs by index when the crop module is not switched on or the crop land unit was not created. We add an ‘iscft’ flag to avoid those hard-wired indices. Flag ‘nfixer’ is used for prognostic crops such as ‘soybean’ and ‘soybeanirrig’, and could be useful for users who may want to implement and/or label non-crop nitrogen-fixing PFTs.

These changes allow, for example, multiple types of shrubs to be defined as separate PFTs that differ from the ELM default of broadleaf_evergreen_shrub, broadleaf_deciduous_temperate_shrub, and broadleaf_deciduous_boreal_shrub. However, the default behavior retains the original ELM PFT definitions so model behavior remains the same when user-defined PFTs are not specifically in use.

References to specific plant functional type numbers throughout the code are changed to use the new flags described above. When using the original ELM PFTs (as defined in the current PFT physiology file format), the updated code will set all the above flags to preserve the original model behavior.

Activating and using user-defined PFT functionality

On model initialization, the updated code checks the PFT names list in the PFT physiology file and compares all PFT names to the default PFT list. If all names and their ordering match those of the default PFTs, and new flags are not present in the physiology file, then flags will be set for all PFTs to preserve the original model behavior. Otherwise, PFT flags will be set using the “woody”, “needleleaf”, “nonvascular”, “crop”, and “evergreen”, “graminoid”, and “iscft” flags defined in the PFT parameters file. In summary, the new, user-defined PFT behavior can be activated by supplying a PFT physiology file with customized PFT names and PFT-specific flag values for the fields listed above.

Note that the total number of PFTs defined in the PFT physiology file must match the number of PFTs in the surface data file. If the number of PFTs is changed from the default (17), the model build configuration must also be changed to reflect the updated number of PFTs. For example, for a model configuration with 12 total PFTs:

```
./xmlchange --id ELM_BLDNML_OPTS --val "-maxpft 12"
```

Dormancy of root respiration:

This code update adds two new parameters:

dormant_mr_temp: Temperature below which dormancy will be activated for fine root maintenance respiration. Default: 0 (K)

dormant_mr_factor: Factor by which maintenance respiration will be scaled when temperature is below the threshold. Default: 1.0 (unitless).

The new code updates root maintenance respiration so when the temperature in a soil layer is below dormant_mr_temp, the temperature factor applied to maintenance respiration is multiplied by dormant_mr_factor. Note that the changes have no effect when the default values are used, so model behavior only changes when a plant physiology file including these new parameters, with different values from the defaults, is used. The parameter values used in arctic simulations were dormant_mr_temp = 272.15 K and dormant_mr_factor = 0.05. The parameters each have a single value that applies to all PFTs (not PFT-specific individual values).

Rhizome allocation for nonwoody plants:

The new code activates allocation and turnover of livecrootc, livecrootn, and livecrootp for nonwoody plants. Allocation to these pools is controlled by the existing stem_leaf parameter, which in the default PFT physiology file is set to zero for nonwoody plants. When the stem_leaf parameter is greater than zero for nonwoody plants, its value is used to allocate carbon to the livecroot pools. N and P allocation are controlled by the existing livewdcn and livewdcp parameters. Note that in the default physiology file livewdcn and livewdcp are set to zero for nonwoody PFTs, and these values need to be changed when using the rhizome functionality. Rhizome turnover time is controlled by a new parameter rhizome_long that can be set with PFT-specific values when using this functionality. If rhizome_long is not supplied in the physiology file, its value will be set to the same as froot_long.

Evergreen plant growth from storage pools:

The fcur parameter controls the amount of photosynthate that is allocated to current growth rather than storage. The default physiology file uses a value of fcur = 1.0 for all evergreen plants, assuming no storage. In the updated code, fcur can be set to values less than 1.0 to allow for some allocation to storage pools. The transfer rate of storage pools to growth is controlled using the existing fstor2tran parameter. Because evergreen plants do not currently have defined phenological stages in the model, this transfer occurs at a constant rate with fstor2tran interpreted as the fraction of storage allocated to growth per year.

PFT-specific nitrogen fixation

The current model uses a hard-coded nitrogen fixation function which calculates column-level nitrogen fixation rate as a function of annual NPP with fixed parameters:

$$N_{fix} = 1.8(1 - e^{-0.003 NPP_{ann}})$$

Where N_{fix} is nitrogen fixation rate ($\text{g N m}^{-2} \text{s}^{-1}$) and NPP_{ann} is annual average NPP at the column level ($\text{g C m}^{-2} \text{s}^{-1}$). In the updated code, the 1.8 and 0.003 parameters are added to the PFT physiology file as PFT-specific parameters. The N fixation calculation is changed to use a weighted average of N fixation calculated for each PFT using the PFT-specific parameters:

$$N_{fix} = \frac{\sum_p C_{1,p}(1 - e^{-C_{2,p} NPP_{ann}})wt_p}{\sum_p wt_p}$$

Where p is the patch (with a specific PFT), C_1 and C_2 are new PFT-specific parameters, and wt_p is the area fraction of each patch. The new parameters are read from the PFT physiology file as $Nfix_NPP_c1$ and $Nfix_NPP_c2$, respectively. If not present, they are given the default values of 1.8 and 0.003. This preserves the original column-level N fixation behavior when not specifically set with an updated physiology file.

Table 1: New flags that will be assigned for default ELM PFTs. All parameters represent values in the `clm_params.nc` file that is processed by 'pftconrd' subroutine in `pftvarcon.F90`.

PFT number	pftname	woody	nonvascular	needleleaf	graminoid	iscft	crop	climate zone
1	not_vegetated	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	needleleaf_evergreen_temperate_tree	1.0	0.0	1.0	0.0	0.0	0.0	2.0
3	needleleaf_evergreen_boreal_tree	1.0	0.0	1.0	0.0	0.0	0.0	3.0
4	needleleaf_deciduous_boreal_tree	1.0	0.0	1.0	0.0	0.0	0.0	3.0
5	broadleaf_evergreen_tropical_tree	1.0	0.0	0.0	0.0	0.0	0.0	1.0
6	broadleaf_evergreen_temperate_tree	1.0	0.0	0.0	0.0	0.0	0.0	2.0
7	broadleaf_deciduous_tropical_tree	1.0	0.0	0.0	0.0	0.0	0.0	1.0
8	broadleaf_deciduous_temperate_tree	1.0	0.0	0.0	0.0	0.0	0.0	2.0
9	broadleaf_deciduous_boreal_tree	1.0	0.0	0.0	0.0	0.0	0.0	3.0
10	broadleaf_evergreen_shrub	2.0	0.0	0.0	0.0	0.0	0.0	0.0
11	broadleaf_deciduous_temperate_shrub	2.0	0.0	0.0	0.0	0.0	0.0	2.0

12	broadleaf_deciduous_boreal_shrub	2.0	0.0	0.0	0.0	0.0	0.0	3.0
13	c3_arctic_grass	0.0	0.0	0.0	1.0	0.0	0.0	4.0
14	c3_non-arctic_grass	0.0	0.0	0.0	1.0	0.0	0.0	0.0
15	c4_grass	0.0	0.0	0.0	1.0	0.0	0.0	0.0
16	c3_crop	0.0	0.0	0.0	0.0	0.0	1.0	0.0
17	c3_irrigated	0.0	0.0	0.0	0.0	0.0	1.0	0.0
18	corn	0.0	0.0	0.0	0.0	1.0	1.0	0.0
19	irrigated_corn	0.0	0.0	0.0	0.0	1.0	1.0	0.0
20	spring_temperate_cereal	0.0	0.0	0.0	0.0	1.0	1.0	0.0
21	irrigated_spring_temperate_cereal	0.0	0.0	0.0	0.0	1.0	1.0	0.0
22	winter_temperate_cereal	0.0	0.0	0.0	0.0	1.0	1.0	0.0
23	irrigated_winter_temperate_cereal	0.0	0.0	0.0	0.0	1.0	1.0	0.0
24	soybean	0.0	0.0	0.0	0.0	1.0	1.0	0.0
25	irrigated_soybean	0.0	0.0	0.0	0.0	1.0	1.0	0.0

Table 2: Parameters used for distinguishing user-defined tundra PFTs. Note that no trees were included in this simulation.

PFT number	pftname	woody	nonvascular	evergreen	graminoid	season_decid	nfixer
1	not_vegetated	0.0	0.0	0.0	0.0	0.0	0.0
2	arctic_lichen	0.0	2.0	1.0	0.0	0.0	0.0
3	arctic_bryophyte	0.0	1.0	1.0	0.0	0.0	0.0
4	arctic_evergreen_shrub_dwarf	2.0	0.0	1.0	0.0	0.0	0.0
5	arctic_evergreen_shrub_tall	2.0	0.0	1.0	0.0	0.0	0.0
6	arctic_deciduous_shrub_dwarf	2.0	0.0	0.0	0.0	1.0	0.0
7	arctic_deciduous_shrub_low	2.0	0.0	0.0	0.0	1.0	0.0
8	arctic_deciduous_shrub_tall	2.0	0.0	0.0	0.0	1.0	0.0
9	arctic_deciduous_shrub_alder	2.0	0.0	0.0	0.0	1.0	1.0
10	arctic_forb	0.0	0.0	0.0	1.0	0.0	0.0
11	arctic_wet_graminoid	0.0	0.0	1.0	1.0	0.0	0.0
12	arctic_dry_graminoid	0.0	0.0	1.0	1.0	0.0	0.0

Examples of simulation results:

Shown below are selected visualizations of ELM simulation results using either the default E3SM PFTs or the updated code and parameterizations. These results used a simulation protocol following Sulman et al. (2021). The simulations represent six tundra plant communities measured as part of the NGEE Arctic project: Dryas lichen dwarf shrub tundra (DLST), Birch-ericaceous lichen shrub tundra (BEL), Alder shrubland (AS), Willow-birch tundra (WBT), Alder savanna (ASV), and Tussock-lichen tundra (TT).

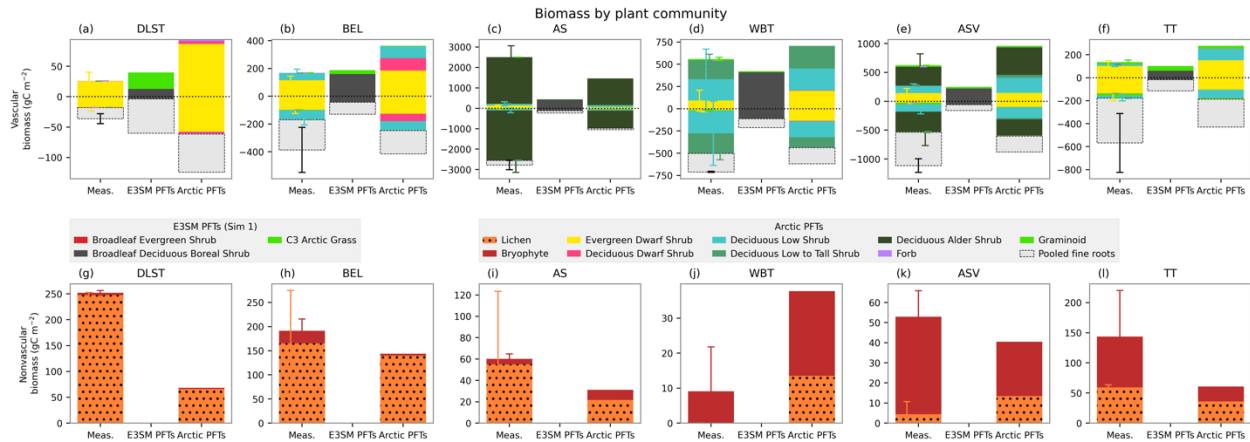


Figure 1: Measured and simulated vegetation biomass. Each bar plot shows measurements (left), simulation results using default E3SM PFTs (middle), and simulation results using new arctic PFTs incorporating new parameterizations and the code changes described above. Different PFTs are shown in different colors, with vascular plant PFTs shown in the top row and nonvascular PFTs in the bottom row. The six panels in each row show different plant communities, following Sulman et al. (2021). Note the more accurate variations in biomass and its distribution across PFTs in the Arctic PFTs simulation, and the higher belowground biomass (shown as negative valued bars).

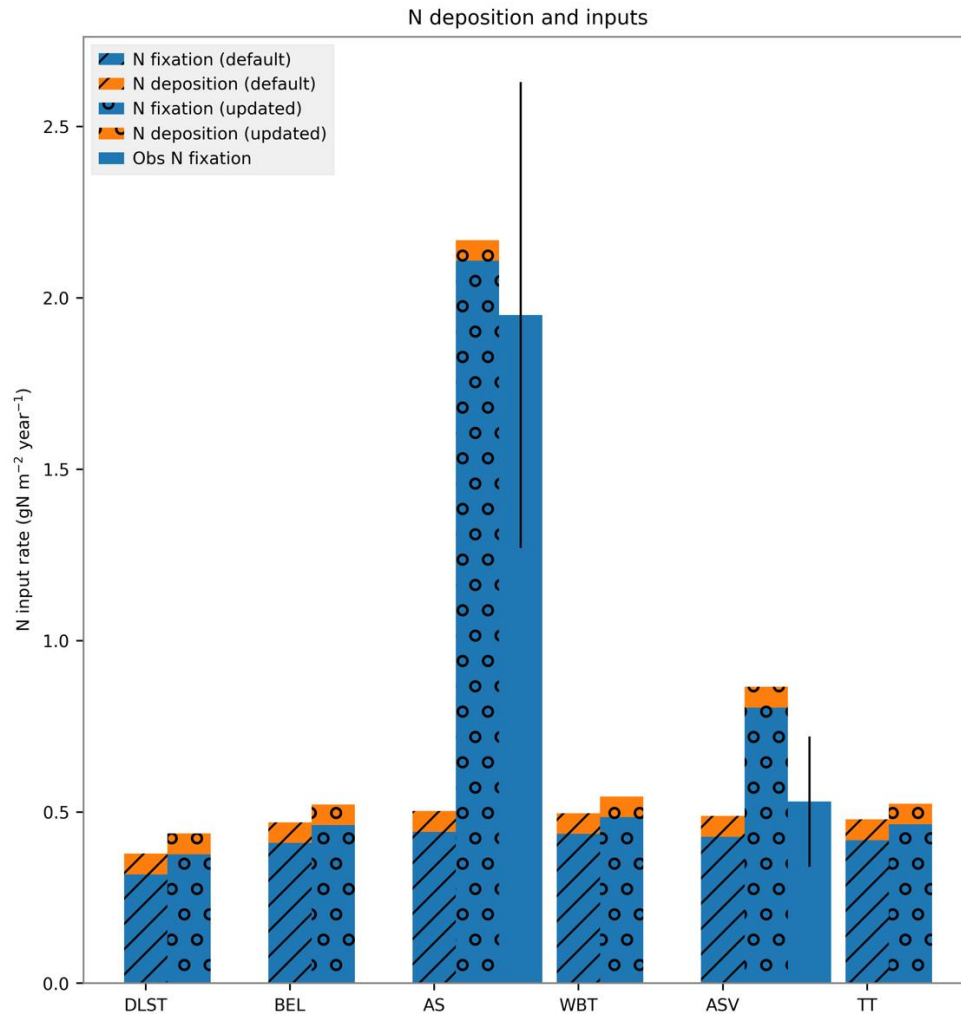


Figure 2: N fixation and N deposition across tundra plant communities, using original (default) E3SM code and modified code that enables PFT-specific N fixation rates. Note the higher N fixation with updated code in the alder shrubland (AS) plant community due to the high fraction of nitrogen-fixing alders, which better matches observed N fixation rates.

Reference:

Sulman, B. N., Salmon, V. G., Iversen, C. M., Breen, A. L., Yuan, F., & Thornton, P. E. (2021). Integrating arctic plant functional types in a land surface model using above- and belowground field observations. *Journal of Advances in Modeling Earth Systems*, 13(4), e2020MS002396. <https://doi.org/10.1029/2020ms002396>